3.3 Rational Exponents – Lesson
MCR3U
-

Intro to Rational Exponents (fraction exponents):

$$\sqrt{x}$$

Powers with a rational exponent of the form $\frac{1}{n}$

A power involving a rational exponent with numerator 1 and denominator *n* can be interpreted as the *n*th root of the base:

Example 1: Evaluate each of the following

a) $8^{\frac{1}{3}}$ **b)** $\sqrt[5]{-32}$ **c)** $-16^{\frac{1}{4}}$

d) $\sqrt[4]{\frac{16}{81}}$ **e)** $(-27)^{-\frac{1}{3}}$

Powers with a rational exponent of the form $\frac{m}{n}$

You can evaluate a power involving a rational exponent with numerator *m* and denominator *n* by taking the *n*th root of the base raised to the exponent *m*:

Example 2: Simplify each of the following powers

a) $\sqrt[5]{y^2}$

b) $\sqrt[3]{x}$

c) $\sqrt{a^{-3}b^{\frac{4}{3}}}$

d) $\sqrt[4]{x^3y^2}$

e) $\frac{\sqrt[3]{x^2y} \cdot y^2}{x^3}$

Example 3: Evaluate each of the following

a) $8^{\frac{2}{3}}$

b) $81^{\frac{5}{4}}$

c) $\left(\frac{49}{81}\right)^{-\frac{3}{2}}$

Apply Exponent Rules

Example 4: Simplify and express answer using only positive exponents

a) $\dfrac{\left(x^{\frac{2}{3}}\right)\left(x^{\frac{2}{3}}\right)}{\left(x^{\frac{1}{3}}\right)}$

b) $\left(y^{\frac{1}{4}}\right)^{2} \times \left(y^{-\frac{1}{3}}\right)^{2}$

c) $\left(5x^{\frac{1}{2}}\right)^{2} \times 4x^{-\frac{1}{2}}$

d) $\dfrac{(m^{-2})^{3}\sqrt{m^{4}}}{m\sqrt{pq^{-3}}}$

e) $\dfrac{(x^{2})^{-4} \cdot \sqrt[5]{y^{3}}}{y\sqrt{x^{-2}y}}$